

Negation in Reasoning Traces: Interpretable Signals of Correctness and Provenance

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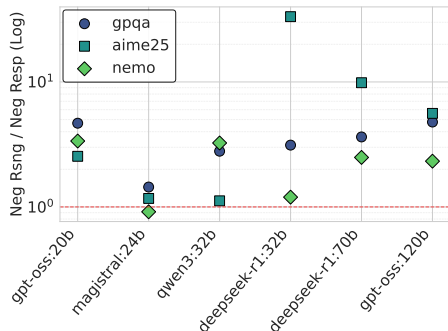
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Motivation: Negation Concentrates in Reasoning Traces

Empirical observation

Across datasets and models tested, negation occurs substantially more often in intermediate reasoning traces than in model inference (roughly $2\times$ to $30\times$ more).



Values above 1 indicate more negation per 100 tokens in reasoning traces than in final-answer text.

Why Discourse-Level Negation?

- Chain-of-thought traces provide a **partial observable record** of intermediate reasoning.
- These traces often show self-monitoring: alternatives are ruled out, conclusions are revised, and hypotheses are narrowed.
- Negation is an explicit linguistic marker of these operations (but not only...).

Research question

Can discourse-level negation in reasoning traces provide interpretable signals of **correctness**, **model identity**, and **human-vs.-LLM authorship**?

From occurrence to function

Simple counts of general negation markers proved too coarse: they indicate that negation is present, but not what role it plays in the reasoning process. We therefore model negation according to its discourse-level function (**how often** does it occur vs. **what it does**).

Refining negation

Narrows the solution space by ruling out alternatives or qualifying an earlier claim.

Corrective negation

Rejects or revises an earlier reasoning step or provisional conclusion.

One Trace, Two Negation Functions

A Illustrative negation examples

Example 1: Discrete Shape Classification

Q: Does the shape of an apple resemble (1) a sphere, (2) a cube, or (3) a torus?

S1: The options are sphere, cube, or torus.

S2: **[Refining]** An apple has **no** sharp edges, so let us rule out the cube.

S3: Then the answer is torus.

S4: **[Corrective]** Wait, an apple has **no** hole, so it **cannot** be a torus.

S5: The apple most closely resembles a sphere.

Neg: S2→S1 (Refining), S4→S3 (Corrective) | **Outcome:** Correct

Example 2: Algebraic Constraint

Q: Solve the equation $x^2 = 25$ given that $x < 0$.

S1: Taking the square root of 25 gives $x = 5$.

S2: So the final answer is 5.

S3: **[Corrective]** Wait, the problem states $x < 0$, so the answer **cannot** be 5.

S4: We must consider the negative root, $x = -5$.

S5: **[Refining]** Are there other negative roots? A positive real number has **no** more than two square roots.

S6: Thus, $x = -5$ is the unique correct solution.

Neg: S3→S2 (Corrective), S5→S4 (Refining) | **Outcome:** Correct

Example 3: Logic / Commonsense

Q: A bat and a ball cost \$1.10. The bat costs \$1 more than the ball.

S1: The total is \$1.10 and the bat is \$1 more, so the ball costs \$0.10 and the bat costs \$1.00.

S2: **[Corrective]** However, \$1.00 is **not** \$1 more than \$0.10.

S3: If the ball cost \$0.10 and the bat cost \$1 more, the total would be \$1.20.

S4: **[Refining]** Therefore, the ball's price **cannot** be \$0.10 or above.

S5: **[Refining]** The price also **cannot** be \$0, as that would make the bat \$1.10 and violate the required difference.

S6: Testing \$0.05 gives a bat price of \$1.05, satisfying both conditions.

Neg: S2→S1 (Corrective), S4/S5→S1 (Refining) | **Outcome:** Correct

Representative annotated reasoning traces illustrating refining and corrective negation.

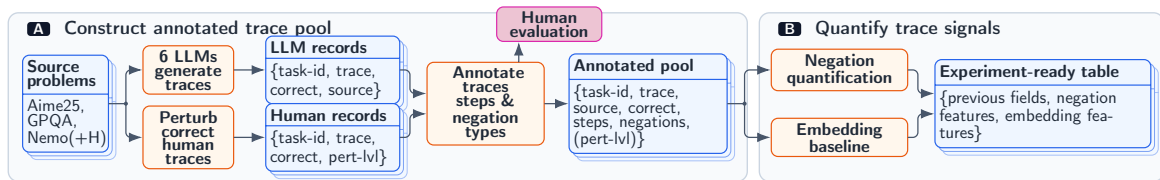
Contributions

- ① **Dataset:** build and partially validate a dataset of human and LLM reasoning traces annotated for refining and corrective negation.
- ② **Metrics:** quantify discourse-level negation through composition, position, and sequential structure.
- ③ **Evaluation:** test whether these features predict correctness, model identity, and human-vs.-LLM authorship.



Data and Annotation Pipeline

- 1 Three base datasets (GPQA, Aime25, Nemotron-Math-HumanReasoning)
- 2 Six models (Gpt:Oss:20b/120b, Magistral:24b, Qwen-3:32b, Deepseek-r1:32b/70b)
- 3 Non-reasoning annotator model: Llama3.3-70b
- 4 Human evaluation from two independent linguists on 36 traces, i.e. 634 reasoning steps.
- 5 Agreement (Kappa) between humans $\kappa = 0.68$, between LLM and humans $\kappa = 0.37$ and 0.38
- 6 Negation metrics + embeddings (MiniLM-L6-v2)



Construction and annotation of the reasoning-trace pool, followed by the extraction of negation-based and embedding-based representations.

Metric Families

Composition (**how much** negation occurs)

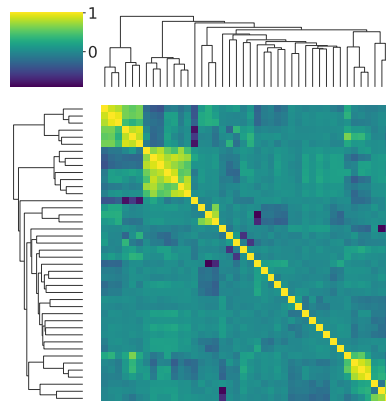
- normalized negation counts
- trace-level negation label
- balance between negation types

Position (**where** negation occurs)

- average normalized position
- first and last negation type
- first- vs. second-half frequency

Structure (**how** negation is **organized**)

- gaps and consecutive runs
- switches between negation types
- randomness and autocorrelation



Hierarchically clustered heatmap of correlations between negation-quantifying metrics.

Experimental Setup

Tasks

- 1 **Correctness prediction**
correct vs. incorrect final answer
- 2 **Model identity**
which LLM generated the trace
- 3 **Human-vs.-LLM authorship**
balanced correct Nemo traces

Controlled extension

Human correctness prediction via perturbing correct Nemo human traces at three severity levels.

Feature settings

- (N) negation features
- (S) structural baseline (length, #steps, #sents...)
- (E) MiniLM-L6-v2 embeddings
- (E + N) embeddings + negation

Classifier and evaluation

- LightGBM gradient boosting with five-fold cross-validation (F1 scores reported as fold averages).

A Experiments

Correctness prediction

Gradient boosting; features → correct / incorrect; human-trace contrast set

Model attribution

Gradient boosting; features → model identity

Human vs. LLM traces

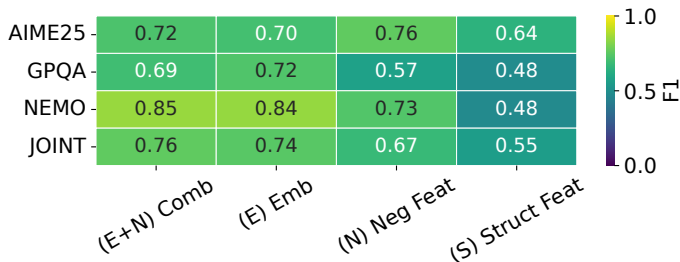
Gradient boosting; features → human / LLM source; restricted to correct Nemo traces

Interpretability
/ SHAP analysis

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graph TD; A[Correctness prediction] --> D[Interpretability / SHAP analysis]; B[Model attribution] --> D; C[Human vs. LLM traces] --> D;
```


Correctness: Negation Beats Simple Structure, Not Embeddings Overall

- Individual negation metrics are weak predictors in isolation.
- As a feature set, negation consistently outperforms the structural baseline.
- Embeddings are generally stronger overall.
- Negation adds small complementary gains in several combined settings.



Negation features improve over the structural baseline by **27.85%** on average across datasets, but cannot beat the embedding baseline.

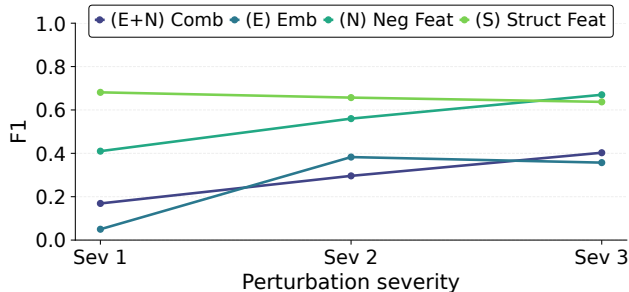
Human Traces: A Controlled Perturbation Stress Test

Why perturb?

Nemo contains only correct human-authored reasoning traces. We generate incorrect variants by injecting increasingly severe human-like reasoning errors.

Interpretation

This is a controlled sensitivity test, not a direct model of naturally occurring human failure.

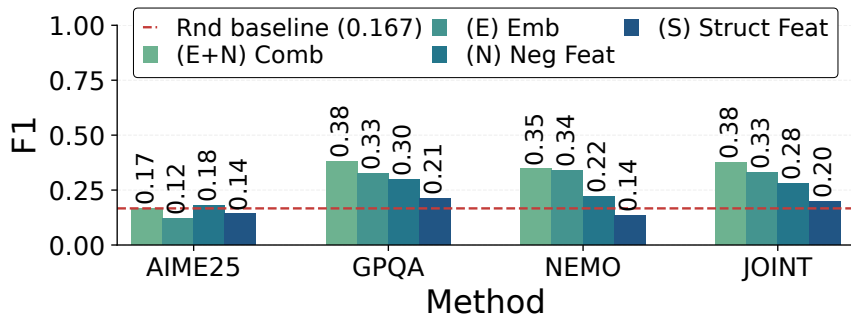


Five-fold CV binary F1 under increasing perturbation severity.

Structural features remain unexpectedly strong but decline with severity; negation features become more discriminative as injected errors grow.

Model Identity: Weak but Complementary Signal

- Predicting the exact LLM from a trace is difficult.
- Random baseline for six models: macro F1 ≈ 0.166 .
- Negation features still outperform the structural baseline across datasets.
- Embeddings are stronger, but adding negation improves them in every setting.



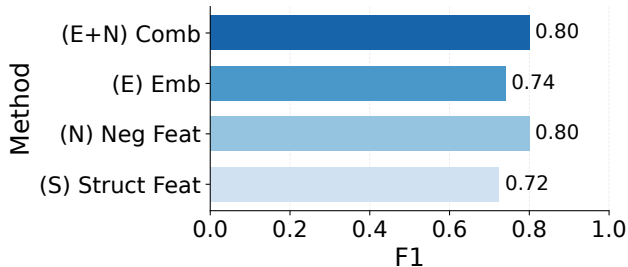
Negation captures modest but real model-specific stylistic signal.

Authorship: Negation Best Distinguishes Human from LLM Traces

Controlled comparison

Balanced setting using correct human and correct LLM traces from the same Nemo dataset.

- Negation features perform best: **.80 F1**.
- Embeddings reach **.74 F1**; structural features reach **.72 F1**.
- Negation + embeddings does not improve beyond **.80 F1**.



Five-fold CV binary F1 for human-vs.-LLM authorship classification.

The strongest result is provenance-related: negation captures a clear difference between human and model-generated reasoning traces.

Interpretability: Different Tasks Use Different Negation Signals

We aggregate normalized mean absolute SHAP values by negation function and metric family. Predictive importance does not imply causal effectiveness.

Importance by negation function

Task	Refining	Corrective
Correctness prediction	64%	36%
Model identity	43%	57%
Human vs. LLM	58%	42%

Mean importance by family

Metric family	Mean
Structural	41%
Compositional	30%
Positional	29%

Correctness relies more on refining-related variation; provenance tasks use a broader mix. Across tasks how negation is sequentially organized has a larger impact on the classification decision, than how often and where it occurs..

Interpretability: Human and LLM Traces Differ in Negation Style

FDR-significant differences

Feature	Human	LLM	SMD
Longest corrective run	1.111	0.306	+.561
Corrective negations per step	0.125	0.051	+.623
Refining negations per step	0.031	0.079	−.476

Positive SMD: higher for humans. Negative SMD: higher for LLMs.

Human traces

More corrective negation per step, with longer repeated runs of corrective revision.

LLM traces

More refining negation per step, suggesting more restriction of alternatives than overt correction.

No positional feature survives group-wise FDR correction: the strongest human–LLM differences concern negation function and sequential organization.

Scope and evidence

- Focuses on **explicit** negation; implicit denial, contrast, or pragmatic opposition are not captured.
- Automatic labels are weak proxy annotations with limited human validation.
- Human incorrect traces are synthetically perturbed, not naturally failed reasoning.
- Evaluation is restricted to textual, CoT-style reasoning traces and a limited task set.

Takeaways

- ➊ Negation is strongly enriched in reasoning traces relative to final responses.
- ➋ Refining and corrective negation capture different forms of self-revision.
- ➌ The strongest result is provenance-related: negation reaches **.80 F1** for human-vs.-LLM authorship.

Negation offers a linguistic lens on how reasoning traces **reject**, **restrict**, and **revise** prior commitments.

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